Large Language Models for Structured Data Extraction from Unstructured Clinical Notes: A Systematic Evaluation

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ABSTRACT

Electronic health records contain vast amounts of clinically valuable information locked in unstructured free-text notes. Manual extraction is impractical at scale; traditional NLP systems require expensive annotation. We evaluate 8 large language models (LLMs) — including both proprietary and open-source variants — on 12 structured extraction tasks drawn from real clinical notes (n=15,000, de-identified). Tasks include medication extraction, symptom coding, adverse event detection, and diagnostic criteria identification. Top-performing models achieve F1 scores of 0.91–0.94 on medication extraction and 0.85–0.89 on symptom coding. Critically, we identify systematic failure modes specific to clinical language — negation handling, temporal reasoning, and rare disease terminology — that are not captured by standard NLP benchmarks.

METHODS

- Dataset: 15,000 de-identified clinical notes from 3 hospital systems (ICU, outpatient, ED)
- Tasks: 12 extraction tasks; annotated by 2 clinical NLP experts + 1 clinician adjudicator
- Models: GPT-4o, Claude Sonnet 4.x, Gemini Ultra, Llama 3 70B, Mistral-Large, BioBERT-Large, ClinicalBERT, GatorTron
- Evaluation: token-level F1, exact match, inter-annotator agreement (Cohen's κ)
- Prompt engineering: zero-shot, 5-shot, chain-of-thought variants all tested

KEY FINDINGS

1. Large general LLMs (GPT-4o, Claude Sonnet) match or exceed domain-specific fine-tuned models on most tasks
2. Negation failures account for 34% of all errors — "patient denies chest pain" misclassified as positive
3. Chain-of-thought prompting improves temporal reasoning by 18%
4. Open-source models (Llama 3 70B) achieve 94% of proprietary model performance at 1/10 cost
5. All models struggle with rare disease terminology (< 1:10,000 prevalence) — F1 drops to 0.61

LIMITATIONS

- Notes from US hospital systems — may not generalise to other languages or healthcare systems
- De-identification may alter note characteristics
- No evaluation of downstream clinical decision making
- Hallucination rate not systematically characterised

RELEVANCE TO ML IN CLINICAL DIAGNOSTICS: HIGH

Critical for anyone considering LLM-based clinical text processing. The systematic failure mode analysis is immediately practical for deployment risk assessment.